

DSCI550: Data Science at Scale
Homework 3, Spring 2026
Due: 11:59pm, Feb. 27, 2026.

SHOW EACH STEP OF COMPUTATION.

1. (10 pts) Overfitting is one of the reasons that may make a machine learning algorithm fail on some learning tasks.
 - a) (5 pts) What is overfitting?
 - b) (3 pts) What is a technique that can help avoid overfitting?
 - c) (2 pts) Other than overfitting, what else can be a reason that may make a machine learning fail.

2. (10 pts) Briefly answer the following questions in one or two sentences.
 - a) (5 pts) What is cross-validation in modeling?
 - b) (5 pts) Classifier A has 90% accuracy on the training set and 70% accuracy on the test set. Classifier B has 78% accuracy on both the training and test sets. Therefore, we can conclude that classifier A is better than classifier B (because it has better mean accuracy). True or False. Explain.

3. (10 pts) (Bayes rule) A patient takes a lab test with two possible results (P, N), and the result comes back positive. It is known that the test returns
 - A correct positive result in only 98% of the cases (true positive, when the tested person has a cancer)
 - and, a correct negative result in only 97% of the cases (true negative, when the tested person has no cancer).
 - Furthermore, only 0.008 of the entire population has this disease.

Using Bayes rule,

 - a) What is the probability that this patient has cancer?
 - b) What is the probability that he does not have cancer?
 - c) What is your diagnosis, cancer or not?

4. (15 pts) (Naïve Bayes Classifier) We have data on 1000 patients. They were diagnosed to be Flu, Allergy, or Other Disease using three symptoms as shown. This is our 'training set.' We will use this to predict the type of any new patient we encounter.

Type	Fever		Sneezing		Runny Nose		Total
	Yes	No	Yes	No	Yes	No	
Flu	400	100	350	150	450	50	500
Allergy	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

A new patient says “High fever, No Sneezing, and Runny Nose”. Is this Flu, Allergy, or Other? Use Naive Bayes Classifier.

5. (15 pts) Regression: A researcher wants to know if there is a relationship between the number of shopping centers in a state and the retail sales (in billions \$) of that state. A random sample of 8 states is listed below. After determining, via a scatter-plot, that the data followed a linear pattern, the regression line was found. Using the given data and the given regression output answer the following questions.

State Num Sales

1	630	15.5
2	370	7.5
3	616	13.9
4	700	18.7
5	430	8.2
6	568	13.2
7	1200	23.0
8	2976	87.3

- What is the equation of the regression line?
- Interpret the slope in the words of the problem.
- Find r^2 and interpret its meaning in the words of the problem.
- Find the error for predicting the sales of a state with 1200 stores.
- Can you properly predict the sales for a state with 100 stores with the regression line?

6. (20 pts) **A Priori:**

- 1) (10 pts) You have the following five transactions from a supermarket.

TID	List of Items
1	A, B, C
2	B, C, D, E
3	A, C, D
4	A, C, D, E
5	A, B, C, D

Suppose that the minimum support requirement is 40%. Using the Apriori

algorithm, find all frequent item sets (2 itemset, 3 itemset, 4 itemset.). Show your work step by step.

- 2) Suppose that the itemset $\{B, C, E\}$ has 30% support, $\{B, C\}$ has 40%, $\{B, E\}$ has 30%, $\{C, E\}$ has 50%, and each of B, C, and E alone has 50% support. When the minimum confidence level for a strong association rule is 70%, answer if the following rule is strong.

- 1) (5 pts) IF one buys $\{B \text{ and } C\}$, THEN also buys $\{E\}$.
- 2) (5 pts) IF one buys $\{B\}$, THEN also buys $\{C \text{ and } E\}$.

7. (20 pts) (Hierarchical Clustering) Five, two-dimensional data points are shown below with their distance matrix, i.e., the symmetric matrix that gives the pairwise distance between any two points.

x	y		1	2	3	4	5
-0.43	1.19	1	0.00	1.23	1.34	1.12	1.24
-1.66	1.19	2		0.00	2.16	2.12	1.14
0.12	-0.04	3			0.00	0.40	1.28
0.28	0.33	4				0.00	1.42
-1.14	0.17	5					0.00

Use the distance matrix to perform the following two types of hierarchical clustering: Single Link and Complete Link distance. Show your results step by step, and also draw a dendrogram. Note: the dendrogram should clearly show the order in which the points are merged.